DIGITIZATION OF 8 inch DOBSONION TELESCOPE AND SOLAR STUDY

Project report submitted for the partial fulfillment of the requirements for the award of the degree

of

Bachelor of Technology

In

Applied Electronics & Instrumentation Engineering

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CERTIFICATE

This is to certify that the project report entitled **"digitization of 8 inch dobsonion telescope and solar study"** submitted by Sujan De is a bonafide work carried out by them under the supervision of Mrs. Moumita Sahoo and it is approved for the partial fulfillment of the requirement of the ***Haldia Institute of Technology, Maulana Abul Kalam Azad University of Technology (MAKAUT)*** for the award of the degree of Bachelor of Technology **(*Applied Electronics & Instrumentation Engineering*)** during the academic year 2024-2025.

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ABSTRACT

Accurate tracking and digitization of celestial objects are critical components of modern observational astronomy, allowing for extended observation periods, higher-quality data collection, and a significantly reduced manual workload. Historically, small- and medium-scale telescopes, particularly those operated by educational institutions or amateur astronomers, have relied heavily on manual tracking methods that limit both precision and usability. In response to this challenge, this project presents the design and implementation of a fully digitized and automated tracking system for an 8-inch Newtonian telescope, aimed at enhancing its observational capabilities across a broad range of celestial targets. The initial motivation behind the project was to improve solar observation, enabling precise, continuous tracking of the Sun to facilitate long-duration studies of solar features such as sunspots and prominences without requiring constant manual intervention. To achieve this, the system was architected around a microcontroller-driven motor control platform employing stepper motors to provide fine-grained motion control of the telescope mount. Central to the system's tracking accuracy is the integration of real-time ephemeris data obtained from the Jet Propulsion Laboratory (JPL) Horizons system, which provides highly accurate positional information for solar system bodies and artificial satellites. This data is processed and converted into real-time azimuth and altitude coordinates, which are used to dynamically adjust the telescope's pointing with a high degree of precision. As the project progressed, its scope was broadened well beyond solar tracking. The system was enhanced to support automated tracking of multiple categories of celestial objects, including major planets such as Jupiter and Saturn, natural satellites like Titan, and even complex artificial targets such as the James Webb Space Telescope (JWST). This extended functionality demonstrates both the flexibility and scalability of the system architecture. The automated telescope is capable of maintaining accurate tracking over prolonged observation sessions, adapting dynamically to changing sky positions with minimal tracking error. The project also emphasizes modularity, with an architecture designed to support future integration of advanced features such as high-resolution imaging, remote access and control via web interfaces, and automated scientific data logging. In addition to its technical contributions, the system serves as an effective educational platform, offering students hands-on exposure to a wide range of interdisciplinary concepts, including celestial mechanics, astronomical instrumentation, embedded systems design, automation, and robotics. Ultimately, this work demonstrates that with a combination of modern computational resources and affordable hardware, it is possible to transform traditional manual telescopes into powerful, flexible instruments suitable for both research and education. The project highlights a cost-effective, adaptable approach to the modernization of small-scale observatories, helping to bridge the gap between amateur and professional astronomical research capabilities.

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CHAPTER 1

INTRODUCTION

INTRODUCTION

Telescopes have long been the cornerstone of astronomical exploration, enabling humankind to observe distant celestial bodies and understand the mechanics of the universe. However, traditional telescopes often require manual intervention for alignment and tracking, which limits observation efficiency, especially during long-duration or solar-focused studies. Manual tracking becomes increasingly challenging when precise alignment with fast-moving or low-elevation objects is necessary. In modern astronomy, even at the amateur or academic level, there is a growing need for automation to improve accuracy, minimize human error, and allow integration with digital tools such as imaging systems and real-time data analysis platforms.

This project addresses that need by implementing a low-cost, modular system for automating the motion of an **8-inch Newtonian telescope**, primarily used for solar tracking. The system uses **stepper motors** and a **microcontroller-based control unit** to automate telescope movement. The positional data required for tracking is sourced in real time from the **Jet Propulsion Laboratory (JPL) Horizons system**, which provides high-accuracy ephemeris data for solar system bodies and artificial satellites.

Although the initial goal of the project was to automate solar tracking, the scope has expanded significantly. The system can now track a variety of celestial objects, including **planets**, **Titan** (Saturn's moon), and even the **James Webb Space Telescope (JWST)**. This not only improves the telescope's utility but also makes it a versatile platform for student learning and research in fields such as **astronomical instrumentation**, **celestial mechanics**, and **automation**.

The successful implementation of this system demonstrates how traditional observing tools can be enhanced with modern technology, making precision astronomy more accessible to academic institutions and amateur astronomers alike.

1.1 MOTIVATION AND OBJECTIVE

Astronomical observation has always been a field that blends scientific curiosity with technological advancement. The Sun, as the closest and most studied star, offers a dynamic laboratory for understanding stellar physics, solar activity cycles, and their impact on space weather. However, observing the Sun—and other celestial objects—requires precise, continuous tracking, which is challenging to achieve manually, especially with telescopes lacking automated systems. In educational institutions and small observatories, many telescopes remain manually operated, limiting the efficiency, accuracy, and scope of observational projects

The motivation behind this project originated from the need to modernize an 8-inch Newtonian telescope used for solar studies in our department. Traditionally operated through manual adjustments, the telescope could not maintain accurate alignment with the Sun over extended periods, resulting in data inconsistencies and frequent interruptions. Moreover, manual tracking is not suitable for observing fast-moving targets or conducting time-sensitive studies of solar events such as sunspots, prominences, and flares. Recognizing this limitation, the idea of digitizing and automating the telescope emerged as a solution to enhance its capability and usability.

As the project progressed, it became clear that the same system could be extended to track a wider range of celestial objects. With the integration of real-time ephemeris data from the JPL Horizons system, the project evolved to include automated tracking of planets, natural satellites such as Titan, and even artificial objects like the James Webb Space Telescope (JWST). This expanded functionality not only improved the scientific potential of the telescope but also created an exciting learning platform for students.

The primary objectives of this project are as follows:

- To design and implement an automated control system for an 8-inch Newtonian telescope using stepper motors and a microcontroller-based platform.
- To integrate real-time celestial coordinates from the JPL Horizons system to enable accurate tracking of the Sun and other celestial objects.
- To enhance the telescope's functionality by enabling tracking of multiple object categories, including planets and artificial satellites.
- To develop a modular and scalable system that can be further upgraded with imaging, remote access, and advanced data logging capabilities.
- To provide a valuable educational tool for students to learn about astronomical instrumentation, automation, and celestial mechanics.

1.2 PURPOSE, SCOPE, AND APPLICABILITY

Purpose:

The primary purpose of this project is to design and implement a cost-effective and modular automation system for an **8-inch Newtonian telescope**, enabling it to track the Sun and other celestial bodies with high precision. Traditionally, such telescopes require manual adjustments to maintain alignment with moving celestial targets. This becomes inefficient and error-prone during extended observation sessions. The project aims to digitize the telescope's motion control using stepper motors and microcontroller-based systems, coupled with real-time ephemeris data from the **Jet Propulsion Laboratory (JPL) Horizons system**. In doing so, the project seeks to enhance observational capabilities, reduce manual effort, and create a platform for future technological integrations such as imaging, remote access, and automated data acquisition.

Scope:

The scope of this project extends beyond mere solar tracking. While the system is optimized for observing the Sun and its features, it has also been configured to track other celestial bodies including **planets**, **moons like Titan**, and **artificial satellites** such as the **James Webb Space Telescope (JWST)**. The system receives celestial coordinates from JPL, processes them into azimuth and altitude values, and adjusts the telescope orientation in real-time. The project involves mechanical integration of stepper motors, software development for coordinate parsing and motor control, and field testing under real observation conditions. Though initially implemented on a Newtonian telescope, the design is scalable and can be adapted to other telescope types or mounts with minimal changes.

Applicability:

This system has broad applicability in **academic institutions**, **amateur observatories**, and **astronomical outreach centers** where budget-friendly but functional automation is needed. It can serve as a platform for **student training**, **basic research**, and **hands-on learning in astronomical instrumentation**. The system's modularity allows for easy upgrades such as the inclusion of CCD or CMOS imaging devices, cloud-based interfaces, and AI-based target recognition. By bridging the gap between manual and high-end automated telescopes, this project provides an accessible step toward modernizing observational infrastructure in educational and semi-professional environments.

1.3 LITERATURE REVIEW

The evolution of telescope automation has paralleled advancements in embedded systems and low-cost electronics. This chapter reviews relevant literature in solar tracking systems, microcontroller-based astronomical automation, and mechanical design innovations used in amateur and research-grade observatories.

In [1], a dual-axis solar tracker using stepper motors controlled via Arduino was presented, demonstrating how basic microcontrollers can be employed for azimuth and altitude movement. The study showed a 30–40% increase in solar capture efficiency, laying foundational work for low-cost solar tracking.

Authors in [2] designed a GPS and RTC-based solar positioning system, combining astronomical equations with real-time geolocation. Their method highlighted the importance of precise timekeeping and location data, which directly aligns with this project's use of the DS3231 RTC and Neo-6M GPS module.

Berry [3] introduced several Arduino-based amateur astronomy projects, including star tracking, dome automation, and telescope interfacing. His book emphasized the importance of accessible, programmable platforms like Arduino and ESP32 for enabling astronomy in educational settings, which reflects the design philosophy of our system.

In [4], the authors implemented an automated equatorial mount using 3D-printed gears and NEMA 17 stepper motors. The study addressed mechanical backlash, torque requirements, and microstepping resolution—key aspects we considered in designing and 3D-printing custom gear sets for our Dobsonian mount using NEMA 23 motors.

Raza and Elahi [5] explored automated sunspot imaging using low-cost CMOS cameras and solar filters, reinforcing the scientific potential of home-built solar telescopes. Though our project does not include imaging, future integration with such modules could yield similar results.

Finally, in [6], the use of Stellarium and NASA's JPL Horizons system for real-time ephemeris fetching was demonstrated for robotic telescope targeting. This directly inspired the celestial coordinate tracking system implemented in our project using Alt-Az conversion and ESP32-controlled motion logic.

Together, these studies establish a clear technical and conceptual foundation for our telescope automation project, validating the approach of using embedded control, open-source software, and 3D printed mechanics in a scalable and modular framework for solar and celestial tracking.

CHAPTER 2
FUNDAMENTALS OF TELESCOPES AND SOLAR ASTRONOMY

FUNDAMENTALS OF TELESCOPES AND SOLAR ASTRONOMY

2.1 OBJECTIVE

The objective of this chapter is to provide a comprehensive understanding of the fundamental concepts related to telescopes and solar astronomy, which form the scientific and technical foundation of the project. It aims to introduce the basic design principles, classifications, and operational characteristics of telescopes, with a particular focus on the Newtonian reflector used in this work. The chapter further explores the importance of solar observations, highlighting key solar phenomena such as sunspots and prominences, and their relevance to both astrophysical research and space weather studies. A background on telescope mounts and the evolution of automated tracking systems is also presented to establish the necessity of digitization in modern astronomical instrumentation. By combining the theoretical and observational aspects of solar astronomy with the technical understanding of telescopes, this chapter aims to equip the reader with the contextual knowledge needed to appreciate the motivation, design, and implementation of the automated solar tracking system developed in the subsequent chapters.

2.2 TELESCOPES

Telescopes have long been essential tools in the exploration of the universe. They enable us to observe celestial bodies far beyond the reach of the naked eye, helping both amateur and professional astronomers understand the nature of stars, planets, galaxies, and other astronomical phenomena. The fundamental purpose of a telescope is to gather light from distant objects and focus it to form a clear and magnified image. Over the years, various types of telescopes have been developed, each with its own advantages and limitations depending on the purpose of observation. There are primarily three types of telescopes: **refracting**, **reflecting**, and **catadioptric**.

Refracting telescopes use lenses to bend and focus light. They are known for delivering high-contrast images and are often used for planetary observations. However, they are prone to chromatic aberration and become increasingly expensive and heavy as the aperture size increases.

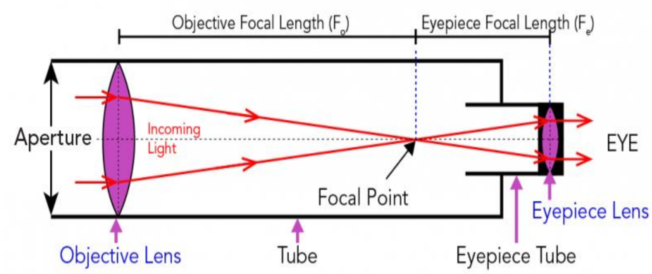

Fig 2.1 : Refracting Telescope

Reflecting telescopes, on the other hand, use mirrors instead of lenses to gather and reflect light. The most well-known design in this category is the **Newtonian telescope**, invented by Sir Isaac Newton. Newtonian reflectors are widely appreciated for their simple construction, cost-effectiveness, and ability to handle large apertures without introducing chromatic distortion. These features make them popular in both amateur and educational setups.

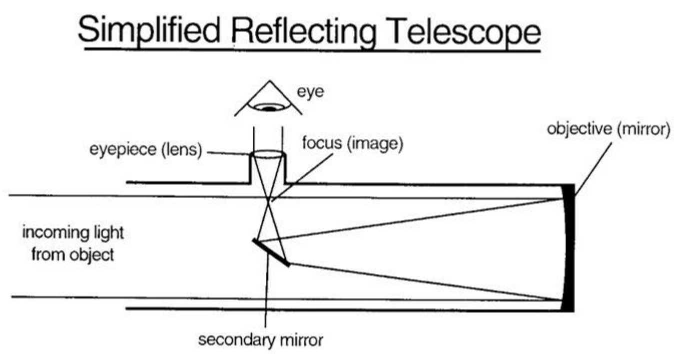

Fig 2.2: Reflecting Telescope

Catadioptric telescopes combine lenses and mirrors to correct for various optical aberrations. While they offer compactness and versatility, they are more mechanically complex and typically costlier.

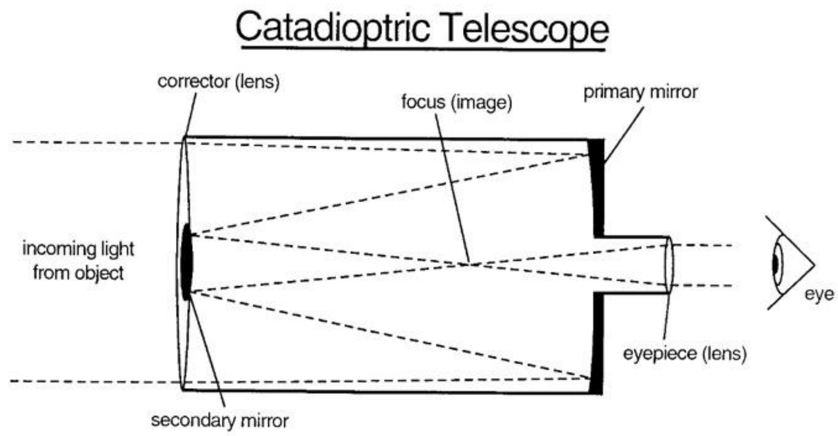

Fig 2.3: Catadioptric Telescope

For this project, we chose an **8-inch Newtonian reflecting telescope** due to its ideal balance between performance, simplicity, and affordability. Its open-tube design makes it easy to adapt with motors and electronic components for automation. The wide aperture allows sufficient light collection for solar and planetary observations, and the reflecting optics ensure minimal distortion. Additionally, the Newtonian design is well-suited for educational and experimental projects, where easy handling, mechanical access, and optical performance are all critical.

Key Optical Parameters

1. **Aperture(D)**
   The **aperture** is the diameter of the telescope's primary objective (lens or mirror), and it directly affects the light-gathering power and resolution.

- **Light-Gathering Power (LGP)** is proportional to the square of the aperture:

$$\mathbf{LGP \propto D2}$$

Thus, doubling the aperture increases light collection fourfold.

2. **Focal_Length(f)**

   The focal length is the distance between the objective and the focal plane where light rays converge. It governs image scale and magnification.

3. **Magnification(M)**

   Magnification is defined as the ratio of the telescope's focal length fff to that of the eyepiece :

$$\mathbf{M = f1/f2}$$

Higher magnification reveals more detail but reduces field of view and image brightness.

4. **Angular_Resolution(θ)**

   This defines the telescope's ability to distinguish between two closely spaced objects. For an ideal optical system (diffraction-limited), the Rayleigh Criterion gives:

$$\boldsymbol{\theta = 1.22\lambda/D}$$

Where:

- θ is in radians,
- λ is the wavelength of light,
- D is the aperture diameter.

5. **Field_of_View(FOV)**

   The field of view depends on the telescope's design and the eyepiece used. It can be approximated as:

$$\mathbf{FOV \approx Apparent\ Field\ of\ Eyepiece/M}$$

2.3 TELESCOPE ANGLES

Altitude and Azimuth Angles

Altitude and azimuth are the two coordinates used in the horizontal coordinate system to locate celestial objects in the sky, relative to an observer's position on Earth.

- *Altitude (Alt) is the angle measured upward from the horizon to the object in the sky.*
  - *It ranges from 0° at the horizon to 90° directly overhead (called the zenith).*
  - *For example, if a star is halfway between the horizon and the zenith, its altitude is 45°.*

- *Azimuth (Az) is the angle measured clockwise from true north along the horizon to the point directly below the object.*
    - *It ranges from 0° to 360°.*
    - *North is 0°, East is 90°, South is 180°, and West is 270°.*

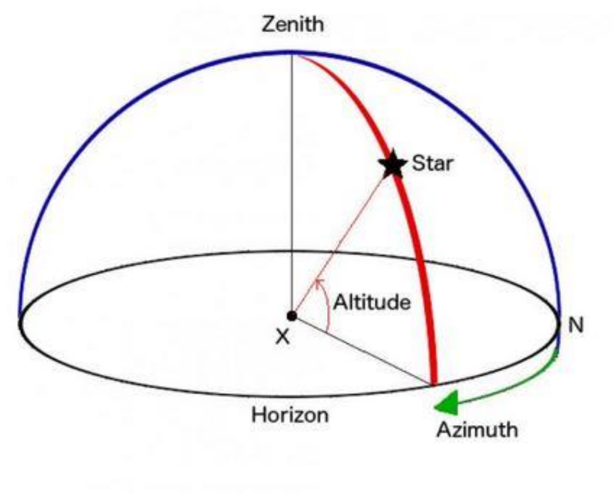

Fig 2.4 : Visualization of Celestial angles used in Telescopes

2.4 TELESCOPE USED IN THE PROJECT

In this project, we have employed the Bresser Messier 8-inch Newtonian Reflector Telescope, mounted on a Dobsonian base. This telescope model was selected for its optimal balance of light-gathering capacity, simplicity of operation, and suitability for modification and automation—key requirements for our solar tracking and digitization objectives.

Key Specifications:

- Optical Type: Newtonian Reflector
- Aperture: 203 mm (8 inches)
- Focal Length: 1000 mm
- Focal Ratio: f/5
- Mount Type: Dobsonian Alt-Azimuth

- Resolution Limit: ~0.57 arcseconds

Fig 2.4: Bresser Messier 8 inch Dobsonion Telescope

2.5 INDRODUCTION TO SOLAR ASTRONOMY

Solar Astronomy is the specialized branch of observational astronomy that focuses on the study of the Sun—our nearest star and the primary source of energy for life on Earth. Unlike other celestial bodies which appear as point sources, the Sun can be observed in great detail due to its proximity, allowing astronomers to study its surface features, atmospheric layers, magnetic activity, and energy output with high temporal and spatial resolution.

Understanding the Sun is not only essential from an astrophysical standpoint but also from a practical and technological perspective. Solar phenomena such as sunspots, solar flares, coronal mass ejections (CMEs), and solar wind directly impact space weather, which in turn affects satellite communications, power grids, GPS systems, and even aviation.

Solar astronomy has traditionally relied on ground-based observatories equipped with specialized filters (like H-alpha and Ca-K) and solar telescopes with solar-safe optics to observe and analyze different layers of the Sun's atmosphere—namely the photosphere, chromosphere, and corona. Modern space missions like SOHO, SDO, and Parker Solar Probe have further enhanced our understanding of solar dynamics through continuous, multi-wavelength monitoring.

In amateur and educational settings, solar observation is typically done using white-light filters placed over the aperture of telescopes, allowing safe viewing of visible features such as sunspots and limb darkening. These features help trace the 11-year solar cycle, track solar rotation, and study the distribution of magnetic activity on the Sun's surface.

In the context of our project, solar astronomy forms the core scientific motivation. Our goal is to build a low-cost, automated tracking system for a Newtonian telescope, capable of following the Sun and other celestial targets using real-time astronomical coordinates. While our current setup does not include imaging instrumentation, it lays the foundation for future integration of cameras and solar sensors to capture and analyze solar activity over extended periods.

By automating solar observation with basic hardware and reliable coordinate systems like those from the JPL Horizons database, this project aligns with the broader vision of digitizing amateur astronomy and making solar studies more accessible and precise in educational environments.

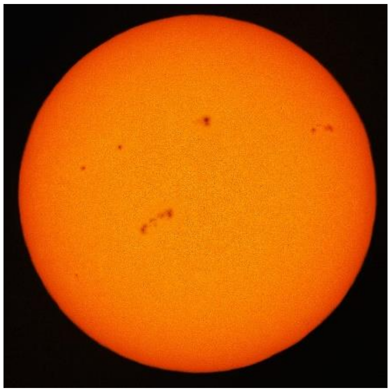

Fig 2.4: Sun Image Taken by Astronomy Outreach club of ARIES, Nainital

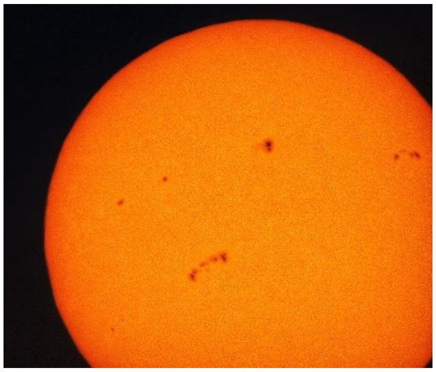

Fig 2.5: Sun Image Taken by Astronomy Outreach club of ARIES, Nainital

CHAPTER 3

TELESCOPE AUTOMATION

TELESCOPE AUTOMATION

3.1 HISTORY OF TELESCOPE AUTOMATION

The automation of telescopes represents a major milestone in observational astronomy, transitioning the field from manual celestial tracking to precise, computer-controlled observation systems. This evolution has been driven by the need for improved accuracy, long-duration tracking, remote operation, and integration with digital data acquisition systems.

Early Manual Telescopes

Historically, telescopes were operated manually using basic mechanical mounts such as **alt-azimuth** or **equatorial** mounts. Observers had to adjust the telescope's position frequently to track celestial objects due to Earth's rotation. Even with accurate star charts, manual operation was time-consuming and limited by human error and physical fatigue.

Introduction of Clock Drives and Mechanical Tracking

The first wave of automation emerged in the form of **clock drives**—mechanical systems powered by weights or motors that rotated the telescope's right ascension axis at **sidereal rate** (~15 arcseconds per second). This allowed continuous tracking of a celestial object across the sky with minimal manual correction. These systems were mostly analog and used in observatories and advanced amateur setups.

Digital Encoders and GoTo Mounts

In the late 20th century, the development of **digital encoders** and **GoTo (Go-To) systems** transformed telescope control. Encoders allowed precise determination of telescope orientation, while GoTo mounts, controlled by microprocessors, could automatically slew the telescope to user-selected celestial coordinates. Users simply selected an object from a database, and the telescope would locate and track it automatically. These systems often included GPS modules, real-time clocks, and databases with thousands of celestial objects.

Integration with Computers and Astronomy Software

With the advent of personal computers and astronomy software such as **Stellarium**, **Cartes du Ciel**, and **NINA**, telescopes could now be controlled via **ASCOM drivers** or **INDI protocols**. This enabled more complex tasks such as scripting, automated observing sessions, time-lapse imaging, and remote operation. Data logging, photometry, and astrometry also became more efficient.

Modern Trends: Arduino, Raspberry Pi, and Internet Integration

In recent years, **DIY automation** using platforms like **Arduino**, **Raspberry Pi**, and **ESP32** has gained popularity, particularly in educational and amateur astronomy projects. These microcontrollers allow for cost-effective automation of telescope movement, tracking, dome control, and even weather monitoring. Web-based interfaces, WiFi connectivity, and real-time coordinate fetching (e.g., from NASA's JPL Horizons system) are making telescope systems more autonomous and accessible.

3.2 THEORY OF TELESCOPE AUTOMATION

Telescope automation refers to the process of mechanizing and digitally controlling the movement and pointing of a telescope without manual intervention. The primary goal is to enable precise and continuous tracking of celestial objects based on their astronomical coordinates, thereby improving observation efficiency and accuracy.

In a typical automated system, a microcontroller (like an ESP32 or Arduino) receives inputs such as time, geographic location, and sensor data (e.g., pitch, roll, and azimuth). Using this information, it calculates the real-time altitude and azimuth (Alt-Az) or right ascension and declination (RA-Dec) of the target object. These coordinates are then converted into motion commands for stepper motors or servo motors that rotate the telescope in both axes.

Telescope automation often involves the use of GPS modules for location, real-time clocks (RTCs) for timekeeping, and IMUs or magnetometers for orientation tracking. These components work together to ensure the telescope remains accurately aligned with the desired object as Earth rotates. Automated systems can also interface with planetarium software and astronomical databases (such as JPL Horizons) to dynamically fetch target positions. This fusion of hardware and software enables not just hands-free operation but also supports remote control, data acquisition, and future integration with imaging systems.

Our automation framework consists of:

- **ESP32 microcontroller** for core processing and control logic.
- **MPU9250 IMU sensor** to measure pitch and roll (aligned with telescope's length axis).
- **QMC5883L magnetometer** for azimuth angle estimation.
- **GPS Module (Neo-6M)** for real-time geolocation.
- **DS3231 RTC Module** for precise time synchronization.
- **TMC2209 stepper motor drivers** to control altitude and azimuth stepper motors.
- **Custom-designed 3D printed gears** integrated with the telescope base and mount.
- **Web-based interface** for manual override and control.

The entire system is powered by a regulated 24V supply, stepped down using DC-DC converters (24V → 5V → 3.3V), as visualized in the schematic and PCB layout.

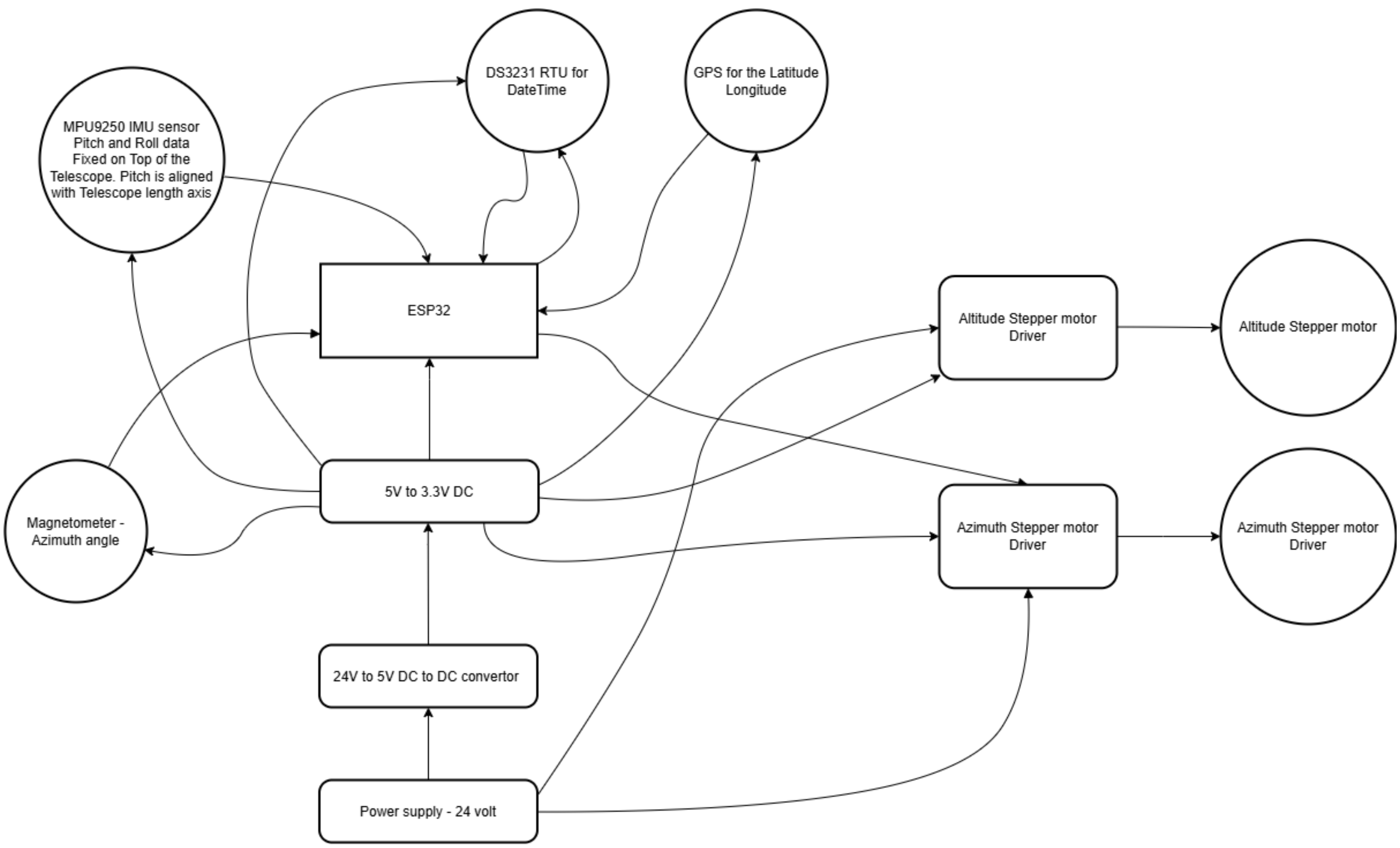

Fig 3.1: Block Diagram of Automation

3.3 ELECTRONIC DESIGN AND INTEGRATION

The telescope automation system is a highly integrated setup that combines sensor data acquisition, real-time processing, and motion control using a microcontroller-based embedded system. The success of the automation depends not only on software and mechanics but heavily on the robustness of the electronic design. This section details the complete hardware architecture, including power management, sensor integration, signal routing, communication interfaces, and PCB development.

3.3.1 System Overview and Architecture

The central control unit of our system is the **ESP32-WROOM-32** microcontroller module, chosen for its dual-core processing capability, integrated Wi-Fi, ample GPIO pins, and built-in UART/I2C/SPI interfaces. It is responsible for reading data from various sensors (IMU, GPS, RTC, magnetometer), computing real-time Alt-Az coordinates, and controlling the stepper motors via TMC2209 drivers.

The system consists of four major electronic domains:

1. **Sensor Interface Subsystem**
2. **Motion Control Subsystem**
3. **Power Management**
4. **Microcontroller Core and Logic Routing**

These domains are integrated and carefully managed through a custom-designed printed circuit board (PCB) to ensure reliability and minimal electromagnetic interference (EMI).

3.3.2 Power Management and Regulation

The system operates from a **24V SMPS external supply**, chosen to provide sufficient headroom for the motor drivers and any external peripherals. The power design includes a two-stage conversion process:

- **Stage 1: 24V to 5V Buck Converter**
  - We used a high-efficiency DC-DC converter module to drop 24V to 5V. This rail powers the motor driver logic and some sensors.
  - Decoupling capacitors (C9, C11, C12) are placed close to the drivers to minimize voltage ripple during high-speed motor operation.
- **Stage 2: 5V to 3.3V LDO Regulator (AMS1117-3.3)**
  - The ESP32 and sensors like the IMU and magnetometer operate at 3.3V. An AMS1117 regulator along with bypass capacitors (C13, C15, C16) ensures stable voltage supply.

Each power rail is isolated using Schottky diodes and protected using ceramic and electrolytic capacitors (100 nF to 1000 µF) to manage both high-frequency and low-frequency noise.

3.3.3 Sensor Integration

a. MPU9250 – Inertial Measurement Unit (IMU)

The MPU9250 provides 9-axis motion sensing: 3-axis gyroscope, 3-axis accelerometer, and 3-axis magnetometer. In our system, we primarily use:

- **Pitch and Roll (X and Y axis accelerometer/gyro)** to determine the tilt of the telescope.
- Mounted on the telescope tube, aligned with the optical axis.
- Communicated over I2C to ESP32 at 400 kHz.
- Connected via pull-up resistors (4.7kΩ) on SDA and SCL lines for signal reliability.

Fig 3.2: MPU9250

b. QMC5883L – Magnetometer

- Provides **Azimuth angle (heading)** of the telescope in real time.
- Connected via I2C and fused with IMU data in software for orientation estimation.
- Faraday cage shielding was added during final implementation to reduce magnetic noise.

Fig 3.3 QMC5883L – Magnetometer

c. NEO-6M GPS Module

- Provides **Latitude, Longitude**, and optional altitude.
- Connected to ESP32 over UART (default 9600 baud).
- Timestamp used to sync observations with RTC.
- Uses an active antenna module to improve satellite fix time (~10-15 seconds cold start).

Fig 3.4 NEO-6M GPS Module

d. DS3231 – Real-Time Clock (RTC)

- Provides **high-accuracy timekeeping (±2 ppm)**.
- Uses temperature-compensated crystal oscillator (TCXO).
- Communicates via I2C.
- Battery-backed; retains time even when the system is powered down.
- Crucial for computing the Local Sidereal Time and Solar/planetary coordinates.

Fig 3.5 DS3231 – Real-Time Clock (RTC)

3.3.4 Stepper Motor Control Subsystem

The telescope movement is handled by two **NEMA 23 stepper motors** mounted on altitude and azimuth axes, respectively.

- Each motor is driven by a **TMC2209 stepper driver**, chosen for:
    - Microstepping support (up to 1/256)
    - StealthChop and SpreadCycle control
    - UART-based configuration (vs traditional DIR/STEP)
    - Current regulation and thermal management
- The drivers are connected to the ESP32 through:

- **STEP/DIR/EN pins** via digital I/O
- **UART pin** for configuring current limit, microstepping, and enabling stall detection

- Heat sinks were added to TMC2209 ICs to prevent overheating during long tracking sessions.

To minimize power usage and heat during idle periods, **motor driver enable signals** are used to cut power when the telescope is stationary for extended periods.

3.3.5 PCB Design and Layout

The custom PCB was designed in **KiCAD** and includes the following key features:

- **Two-layer layout** with power and ground.
- Careful separation between logic and power routing to reduce interference.
- Decoupling capacitors placed within 5mm of IC power pins.
- 45° trace angles and curved corners to reduce signal reflection.
- Wide traces (≥50 mil) for high-current motor paths, especially for VMOT and GND to/from drivers.
- Proper silkscreen labeling of all headers, pinouts, and test points.

Connector placement was optimized based on component physical layout and cable routing to ensure minimal wire crossing and ease of debugging.

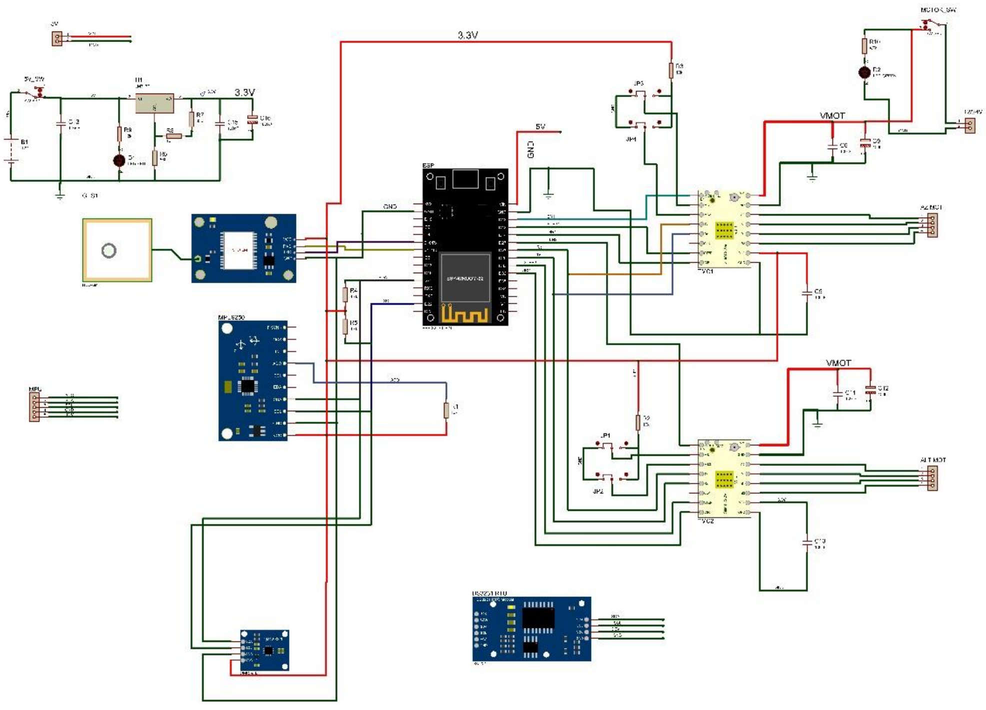

Fig 3.6: PCB Schematic made in KiCAD

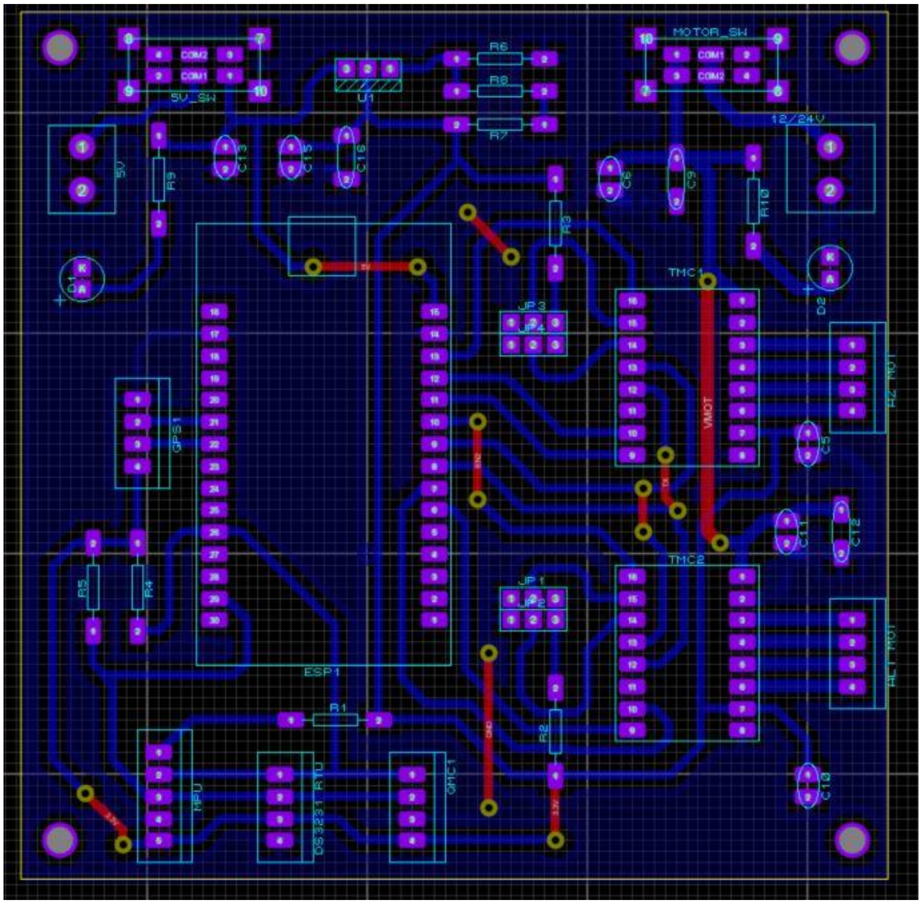

Fig 3.7 PCB Design and Routing

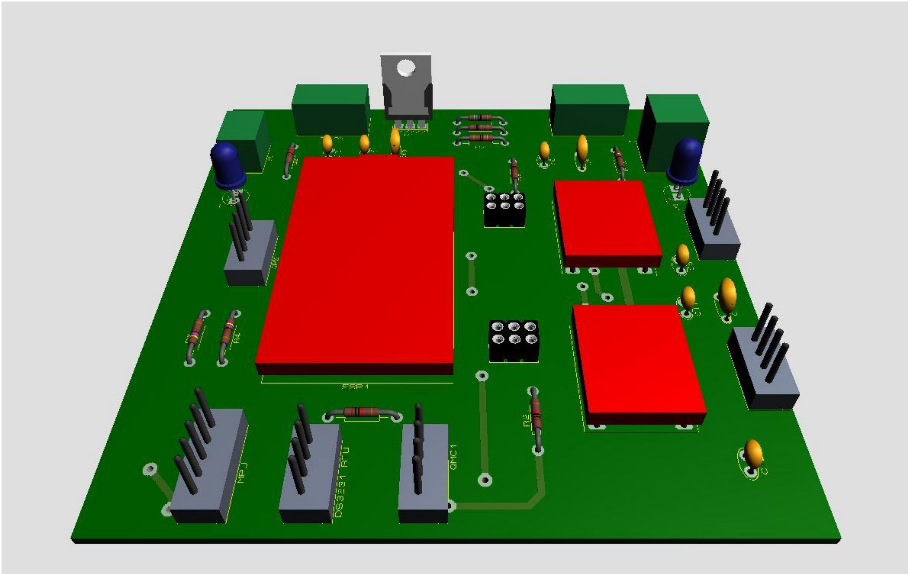

Fig 3.8 : 3D View in KiCAD

3.3.6 Debugging and Expansion

To ensure ease of development and future upgrades:

- *UART and I2C breakout headers were added for logic analyzers or oscilloscope probing.*
- *Dedicated debug LEDs show power status and fault detection.*
- *GPIO headers allow easy firmware re-flashing and boot selection.*

3.3.7 Grounding and Signal Integrity

Ground loops and EMI can cause significant issues in motor control and sensor data. We followed several best practices:

- *Single-point grounding for analog and digital grounds.*
- *Shielded cables for IMU and GPS to reduce EMI from motors.*
- *Used a star-ground topology from the power source to all submodules.*
- *Bypass capacitors and Ferrite beads were added in the final prototype near power entry points.*

3.4 PCB Fabrication Process (DIY Etching Method)

To prototype and integrate the electronics for our telescope automation system, a custom-designed Printed Circuit Board (PCB) was fabricated using the DIY ferric chloride etching method. This process allowed for a cost-effective and quick fabrication of the circuit on a single-sided copper clad board, well-suited for development and testing purposes.

3.4.1 Materials and Tools Used

- *Single-sided copper clad board (FR4, 1.6 mm thickness)*
- *Laser-printed PCB layout on glossy photo paper*
- *Clothes iron for heat transfer*
- *Ferric Chloride ($FeCl_3$) as the etchant*
- *Plastic tray and gloves for safety*
- *Drill machine for component holes*
- *Fine sandpaper and acetone for board preparation*
- *Multimeter for continuity testing*

3.4.2 Design and Printing

The PCB layout was designed using KiCAD and exported as a PDF in monochrome format (black traces, white background). The design was printed using a laser printer onto glossy photo paper. It is essential to use a laser printer because the toner acts as a resist during the etching process, while inkjet printers do not produce the same result.

3.4.3 Copper Board Preparation

1. *The copper clad board was cut to the required dimensions using a hacksaw or PCB cutter.*
2. *The copper surface was cleaned with fine sandpaper or steel wool to remove oxidation and improve adhesion.*
3. *The board was then wiped clean using acetone or isopropyl alcohol to remove any grease or dust.*

This step is crucial to ensure uniform toner transfer during the ironing stage.

3.4.4 Toner Transfer Process

1. *The printed layout (toner side down) was aligned on the copper board.*
2. *A clothes iron set to high temperature (no steam) was pressed over the paper for 5–8 minutes. Firm and even pressure ensured the toner fused well onto the copper.*
3. *After ironing, the board was allowed to cool down, and then the paper was soaked in warm water to gently peel off the photo paper.*
4. *Any residual paper was carefully scrubbed with fingers or a toothbrush.*

This process successfully transferred the toner traces onto the copper board, forming a resist mask.

3.4.5 Etching Using Ferric Chloride

1. *A solution of Ferric Chloride ($FeCl_3$) was prepared in a plastic tray. The solution was slightly warmed using hot water to speed up the etching process.*
2. *The board was immersed in the $FeCl_3$ solution and agitated gently every few minutes.*
3. *After 20–30 minutes, all exposed copper (not covered by toner) was dissolved, leaving behind only the*

toner-protected tracks.

4. *The board was removed and washed thoroughly under running water.*

3.4.6 Cleaning and Drilling

- *The toner was removed using acetone, revealing clean copper tracks.*
- *A 0.8 mm PCB drill bit was used to drill component holes at the designated pads.*
- *After drilling, all holes and pads were cleaned to remove any remaining debris.*

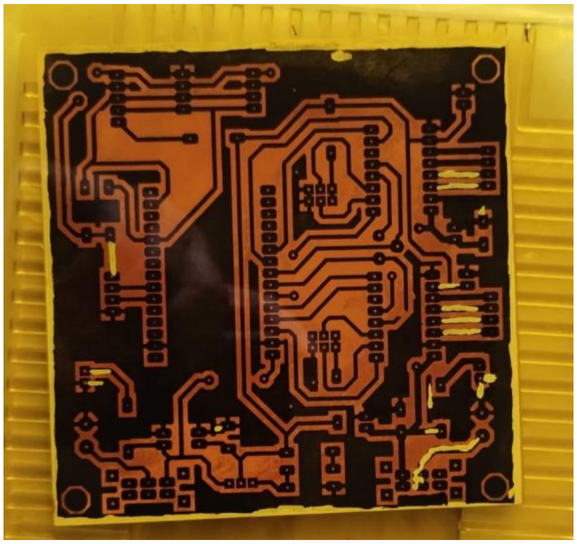

Fig 3.4.1 : Toner Transfer

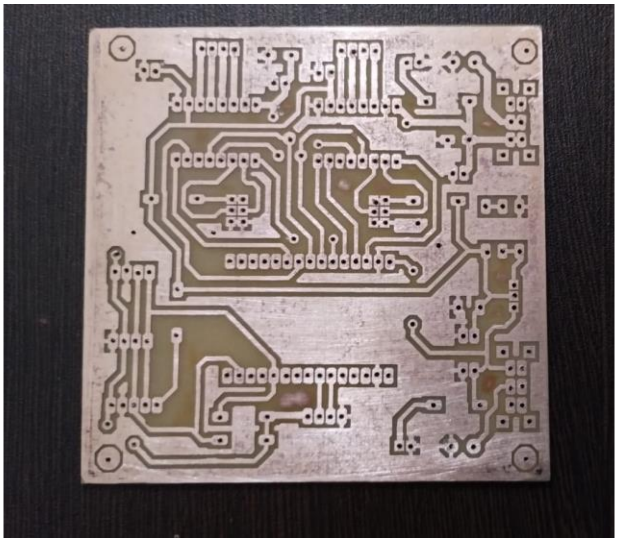

Fig 3.4.2 : Ferric chloride reaction and washing

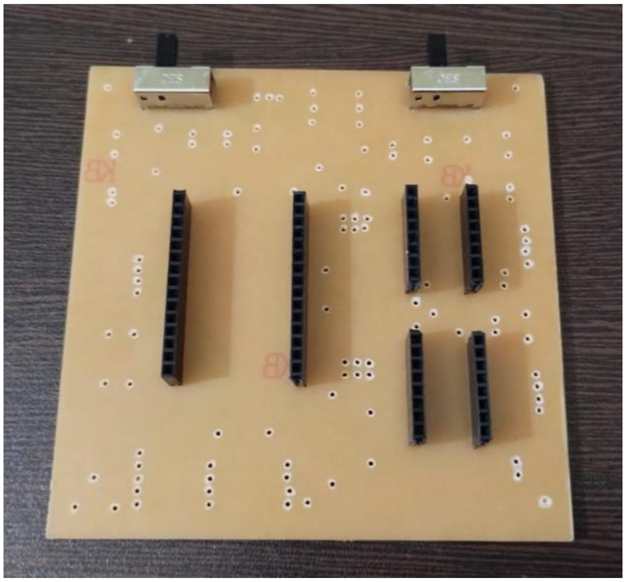

Fig 3.4.3: Drilling and connector placing

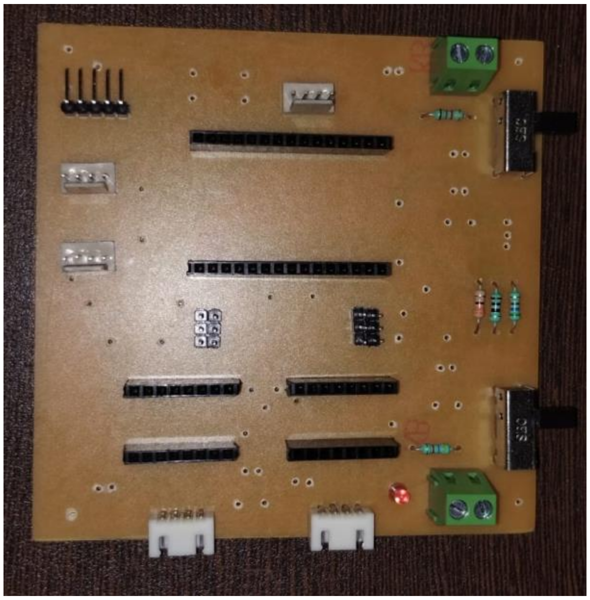

Fig 3.4.4 : Components placing

3.4.7 Inspection and Testing

A visual inspection and a continuity check using a multimeter were performed to ensure all tracks were intact and there were no shorts or breaks. Minor scratches or etching issues were patched with thin copper wire or solder bridges.

The attached image (Figure 3.3) shows the final etched and cleaned PCB board, ready for component soldering.

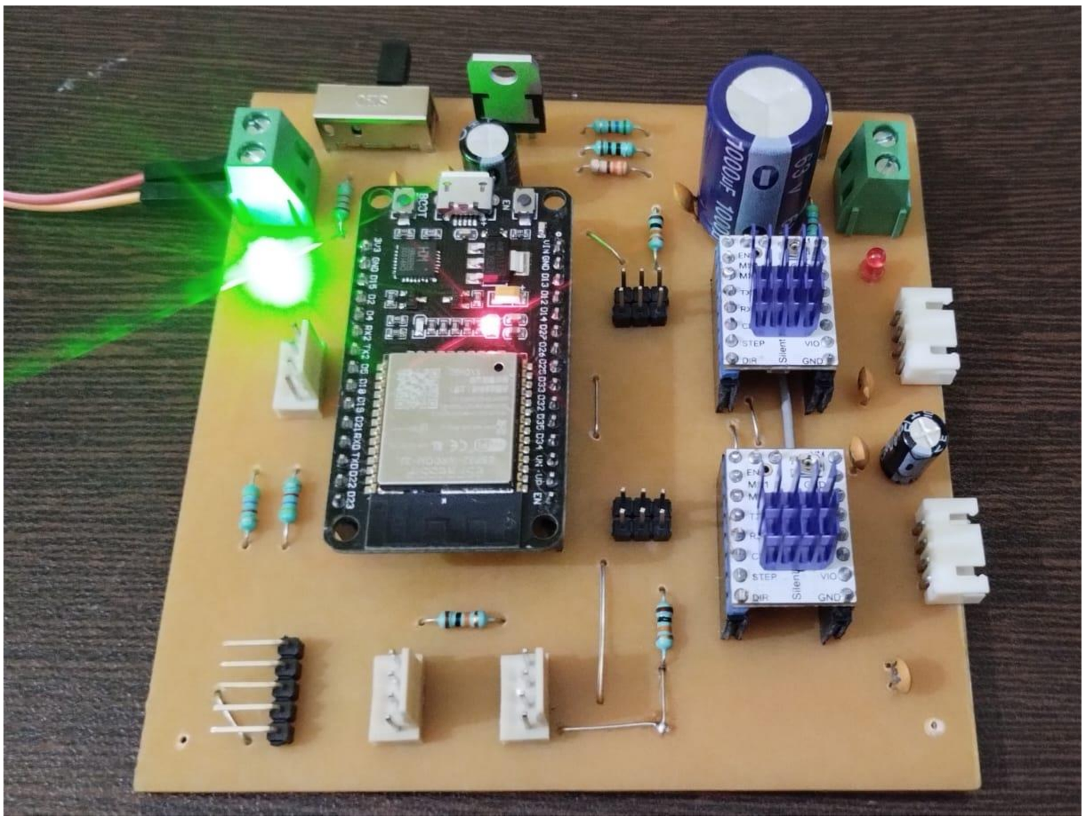

Fig 3.4.5 : Final PCB Board

3.4.8 Remarks on Fabrication Outcome

The PCB was successfully fabricated using the DIY ferric chloride method with well-defined and sharp traces. Despite minor under-etching in a few places, the board performed satisfactorily after manual corrections. This method proved economical and efficient for prototyping purposes, especially in an academic setting without access to professional fabrication facilities.

CHAPTER 4

SOFTWARE INTEGRATION AND MECHANICAL IMPLEMENTATION

SOFTWARE IMPLEMENTATION

4.1 Microcontroller Programming and Software Architecture

This section explains the coding strategy used for automation.

Microcontroller Platform:

- *We used an ESP32 WROOM module due to its high processing power, built-in Wi-Fi, and multiple GPIOs.*
- *It was programmed using the Arduino IDE.*

Libraries Used:

- *Wire.h – for I2C communication.*
- *TinyGPS++ – to parse GPS data.*
- *MPU9250_asukiaaa or MPU6050 – for IMU data.*
- *QMC5883LCompass – for magnetometer readings.*
- *AccelStepper – to control stepper motors.*
- *ESPAsyncWebServer – to host a local control interface.*
- *WiFi.h – for wireless communication and RTC updates.*

Functional Modules in Code:

1. ***Sensor Initialization:***
    - *IMU (MPU9250) and magnetometer (QMC5883L) initialize during boot.*
    - *GPS module sends NMEA strings which are parsed into latitude, longitude, and time.*
2. ***Real-Time Clock (RTC):***
    - *The DS3231 module is used as a timekeeping device.*
    - *If GPS time is available, the RTC is synced on boot.*

3. ***Web Interface:***

   - *A web interface hosted on the ESP32 lets users set microstepping, motor speed, or manually control telescope movement.*

4. ***Stepper Motor Control:***

   - *Two TMC2209 drivers control ALT (altitude) and AZ (azimuth) stepper motors.*

   - *Speed, direction, and microstepping are configurable in real-time via the web interface.*

5. ***Automated Targeting:***

   - *The system calculates the altitude and azimuth of celestial bodies using GPS coordinates and time, then adjusts motor angles accordingly.*

4.2 Stepper Motors and Motor Drivers

To achieve precise control over the telescope's movement in both azimuth and altitude axes, we utilized two NEMA 23 stepper motors, each selected based on the mechanical load and torque requirements of its respective axis.

Stepper Motor Configuration

- ***Azimuth_Axis(Horizontal_Movement):***
  The azimuth rotation is responsible for the horizontal sweep of the telescope. This movement experiences relatively less mechanical resistance, allowing us to use a NEMA 23 stepper motor with 18 kg·cm torque. This motor delivers sufficient holding torque and stepping precision for smooth rotational tracking without excessive current consumption or heating.

- ***Altitude_Axis(Vertical_Movement):***
  The altitude control lifts and tilts the optical tube of the telescope, demanding higher torque to counter gravitational load. To meet this requirement, we used a NEMA 23 stepper motor with 30 kg·cm torque. This motor ensures robust lifting capacity and excellent positional stability during long observation sessions.

Motor Drivers

Both motors are driven by TMC2209 stepper motor drivers, known for their advanced features such as:

- *Ultra-silent StealthChop2 mode for quiet operation*
- *Microstepping support up to 1/256 for high positional accuracy*
- *UART-based configuration for real-time tuning via ESP32*
- *Current control with automatic stall detection and protection*

These drivers were interfaced with the ESP32 microcontroller and received direction, step, and enable signals through GPIO pins. The drivers were mounted on a custom-designed PCB with proper heat sinks and decoupling capacitors to ensure stable operation.

Fig 4.1: Stepper Motor NEMA 23(18kgcm)   Fig 4.2: Stepper Motor NEMA 23(30kgcm)

4.3 3D Printing and Mechanical Assembly

Gear Design in SolidWorks:

- Custom gear profiles for both ALT and AZ axes were designed in **SolidWorks**.
- Gear ratios were chosen to balance speed and torque. For instance, **1:18 gear ratio** provides fine control for celestial tracking.

Printing Parameters:

- Printed using **PLA+ filament**.
- Printer used: Creality Ender 3.
- Infill: 100% for strength.
- Supports: Enabled for overhangs in gear teeth.

Mounting and Integration:

- Gears were fixed to stepper shafts using **grub screws and shaft collars**.
- A rigid support platform was constructed using **aluminum rails and 3D printed holders**.
- Care was taken to reduce mechanical play and alignment errors.

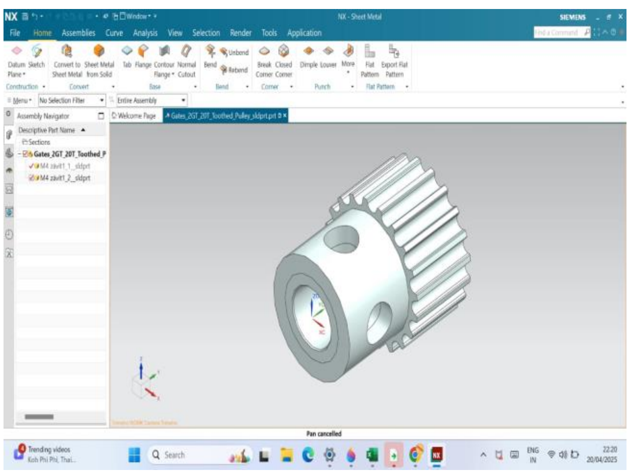

Fig 4.3: Pinion gear for Azimuth angle

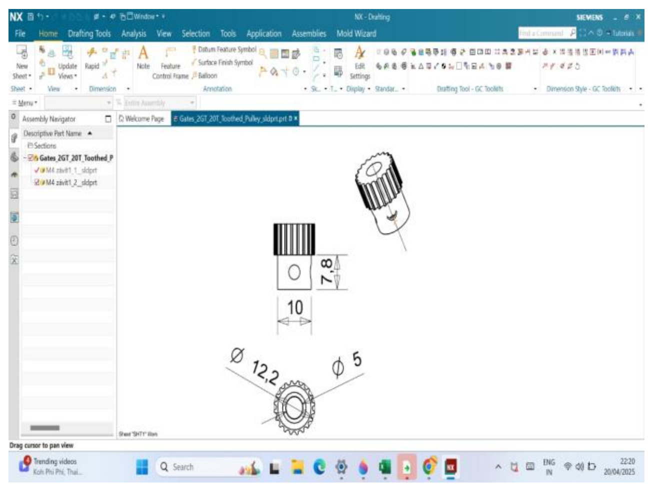

Fig 4.4: Gear dimensions

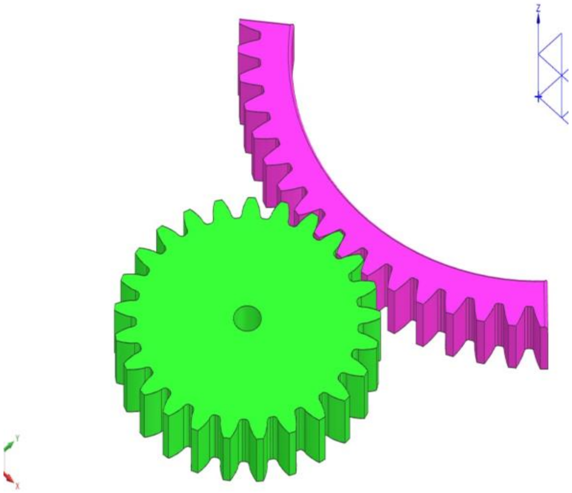

Fig 4.5: Segment Gear and Pinion Gear

Fig 4.6: Final printed pinion gears

Advantages of 3D Printing:

- Rapid prototyping.
- Cost-efficient production.
- Easy customization and replacement.

CHAPTER 5

RESULT ANALYSIS

5.1 Observations

Observation-1

*The image below was captured during solar observation using the automated telescope system developed in this project. The photograph clearly displays the **Sun's disc** through the optical system, taken on **10 June 2025 at 3:48 PM IST**. The observation was conducted under clear sky conditions, and the solar image was focused and tracked using the custom-built **alt-azimuth mount driven by NEMA 23 stepper motors**.*

Fig 5.1.1: Image from our Telescope

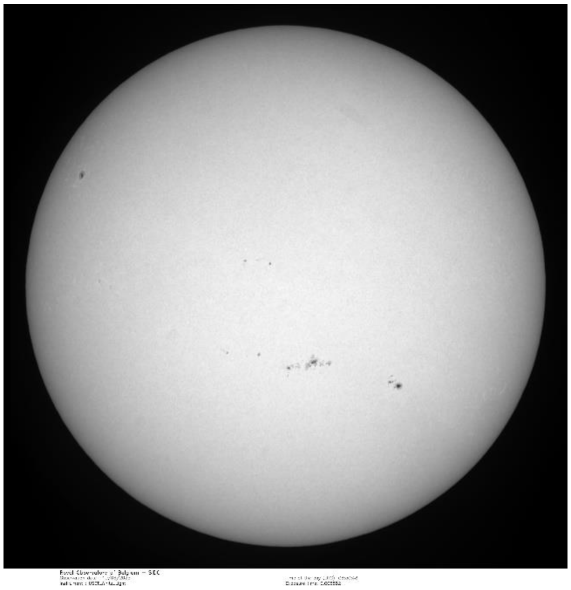

Fig 5.1.2: Image from Royal Observatory of Belgium

Optical Performance

The solar disc is distinctly visible, and the ***circular symmetry*** *of the image indicates good alignment of the telescope optics and successful centering. Minor surface features, possibly* ***sunspots****, are noticeable as tiny dark regions near the center of the disc. These are indications of* ***solar activity****, particularly regions of intense magnetic disturbance.*

Key points regarding image quality:

- ***Uniform Illumination****: The solar disc shows an even brightness profile, confirming the optical axis is properly aligned and that no significant vignetting or optical aberration is present.*

- ***Color and Filtering****: The orange hue indicates that a* ***solar filter*** *(likely ND or Baader film) was effectively used, ensuring observer safety and image clarity by blocking over 99.999% of solar radiation.*

- ***Focus Precision****: The soft but well-defined boundary of the solar disc indicates accurate focusing, which is critical for solar imaging.*

Observation-2

The image below was captured during a solar tracking session on ***6 June 2025*** *using the same custom-built telescope system under partially clear sky conditions. The* ***orange solar disc*** *is clearly visible and covers a significant portion of the field of view. This particular observation showcases the* ***edge of the Sun's disc****, enabling closer inspection of the limb and surface features.*

Fig 5.2.1: Image from our Telescope

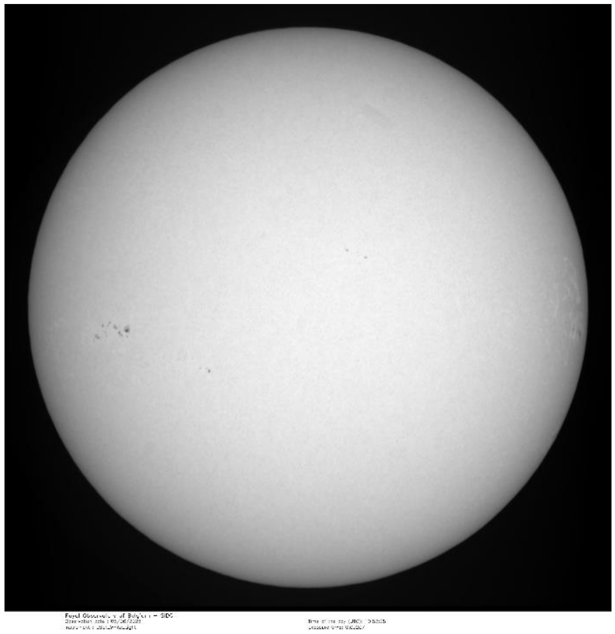

Fig 5.1.2: Image from Royal Observatory of Belgium

Optical Highlights

- ***Limb Observation****: Unlike the previous full-disc capture, this image offers a partial view focused toward the* ***solar limb*** *(the curved outer edge of the Sun). This provides an excellent opportunity to study edge sharpness, chromatic dispersion, and the effect of optical alignment on off-center focusing.*

- ***Sunspot Visibility****: Multiple* ***small sunspots*** *are discernible near the edge of the disc, appearing as fine dark patches. Their presence confirms the telescope's optical resolving power and the current heightened activity phase of* ***Solar Cycle 25****.*

- ***Contrast and Light Scatter****: A slight soft glow around the Sun's edge may indicate* ***internal reflection*** *or atmospheric haze during the capture. However, it adds aesthetic value and demonstrates the scope’s ability to handle high dynamic range without sensor blooming.*

Observation-3

*This image marks one of the earliest successful captures using the automated solar telescope system and represents a significant milestone in the development and testing phase of the project. The observation showcases a relatively calm solar disc with consistent focus and minimal external interference. This image was taken on **2nd May 2025** .*

Fig 5.3.1: Image from our Telescope

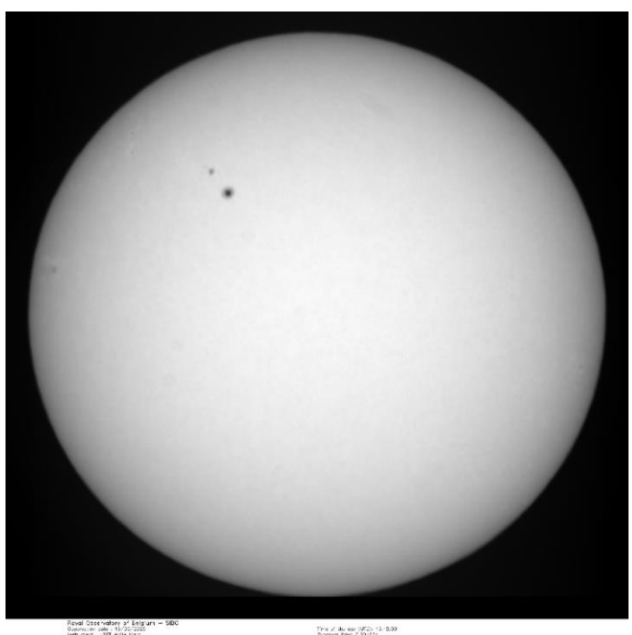

Fig 5.3.2: Image from Royal Observatory of Belgium

Optical and Visual Features

- ***Disc Clarity****: The solar disc is sharply defined, with a* ***smooth and even orange glow*** *across the visible hemisphere. This demonstrates the effectiveness of the* ***solar filter*** *used, which attenuates sunlight intensity to allow safe and observable detail.*

- ***Surface Contrast****: There is a slight gradient visible in the lower half of the disc, possibly caused by* ***lens glare or internal reflections****, which adds a dynamic visual profile without distorting the disc's overall clarity.*

- ***Limb Definition****: The edges of the disc appear clean and rounded, suggesting proper* ***focus and alignment*** *of the optical axis.*

5.3 Tracking and Stability Analysis

In this project, a dual-axis automated solar tracking system was developed using NEMA 23 stepper motors—one rated for ***18 kg-cm torque*** *on the azimuth axis and another rated for* ***30 kg-cm torque*** *on the altitude axis. These were mechanically coupled through* ***custom 3D-printed gears*** *and driven via an Arduino-based motion control algorithm. This section evaluates the real-world performance of the tracking system based on solar observations taken on* ***2nd May, 6th June, and 10th June 2025****, with an emphasis on both strengths and practical challenges.*

Observation 1: 2nd May 2025, 3:22 PM IST

- ***Tracking Status****: Initial tracking was only partially successful. The solar disc appeared within the field of view but was not centered.*

- ***Manual Correction****: The telescope had to be manually adjusted to reposition the Sun to the center during observation.*

- ***Stability****: Despite imperfect tracking, the system maintained enough structural stability for observation and imaging.*

- ***Remarks****: As one of the first test sessions, this helped identify offset errors in alignment and timing.*

Observation 2: 6th June 2025, 3:38 PM IST

- ***Tracking Status****: The system was able to keep the Sun in the general vicinity of the field of view but showed drift over time.*
- ***Manual Correction****: Periodic manual corrections were required to recenter the image.*
- ***Improvements Noted****: Slight refinement in motor speed tuning showed some progress, but overall precision was still not sufficient for long-duration tracking.*
- ***Remarks****: Internal reflections and focus issues persisted, but solar features were more distinguishable.*

Observation 3: 10th June 2025, 3:48 PM IST

- ***Tracking Status****: Marked improvement was observed, with the Sun appearing closer to the center initially.*
- ***Manual Correction****: Still required manual fine-tuning to achieve perfect centering.*
- ***Stability****: Mechanical vibrations were minimal, indicating sound gear engagement, though slight backlash might have contributed to tracking lag.*
- ***Remarks****: This session demonstrated the most stable system behavior, and manual intervention was comparatively lower.*

Overall Performance Summary

Date	***Tracking Precision***	***Manual Correction Needed***	***Image Centering***	***Remarks***
2nd May	*Low*	*Yes (frequent)*	*Off-center*	*First test session*
6th June	*Moderate*	*Yes (periodic)*	*Near-center*	*Mid-phase improvements*
10th June	*Moderate-High*	*Yes (minimal)*	*Almost centered*	*Best performance observed*

5.2 Control Panel Interface and Manual Homing Procedure

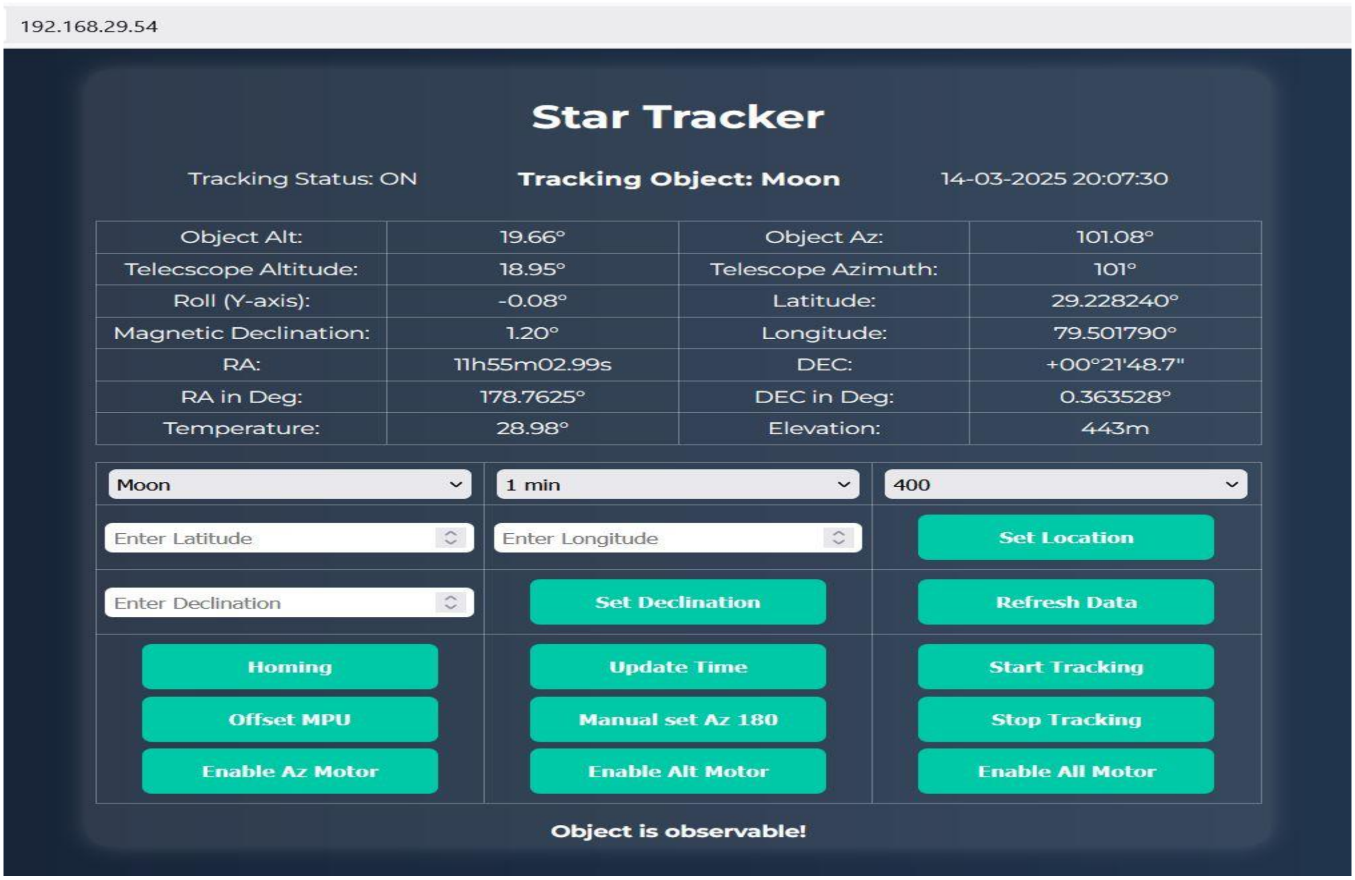

Fig 5.4: Star Tracker Interface and Control Panel

The Star Tracker control panel, accessible via a local IP address (e.g., 192.168.29.54), is a web-based user interface designed to monitor and control the tracking system of an automated telescope. The panel provides real-time astronomical and system parameters, motor control options, and calibration features that are essential for accurate celestial tracking.

Interface Overview

The interface is structured into several key sections:

- *Tracking Status & Object Info: Shows whether tracking is ON or OFF, and the currently selected celestial object (e.g., Moon), with the current date and time.*
- *Live Data Table: Displays real-time parameters including:*
  - *Object and telescope altitude/azimuth*

- *Roll angle of the system*
- *Magnetic declination*
- *RA (Right Ascension) and DEC (Declination) in both HMS and degree format*
- *Temperature, elevation, latitude, and longitude*

- *Object Selection and Time Interval: Dropdown menus allow the user to select the tracking object, observation duration, and motor speed (steps or RPM).*
- *Manual Input Fields: Latitude, longitude, and declination can be manually entered and set using the respective buttons.*

Manual Homing Procedure

Before starting automatic tracking, it is important to home the telescope manually so the system's reference orientation aligns with the actual sky coordinates. Here's the step-by-step procedure:

1. ***Manual Set Azimuth to 180°:***
    - *Click the "Manual set Az 180" button to rotate the azimuth motor and align the telescope to the 180° mark (South direction).*
    - *Use a physical compass to ensure the telescope is facing true South.*
2. ***Manual Set Altitude to 180°:***
    - *Adjust the altitude by clicking a button or physically moving the telescope until it reaches 180°.*
    - *This sets the reference point for vertical orientation.*
3. ***Offset MPU (Inertial Sensor):***
    - *After the telescope is physically aligned to South and vertically leveled, click "Offset MPU" to calibrate the inertial measurement unit (IMU) and magnetometer with the current orientation.*
4. ***Homing:***
    - *Press the "Homing" button to save this orientation as the initial position (home) for the system.*

- *This synchronizes the software's virtual model of the telescope position with its real-*world *orientation.*

Once homed, the telescope can begin accurately tracking celestial bodies using real-time data fetched from JPL Horizons or internal astronomical calculations.

5.4 Limitations and Future Scope

Despite the success of this observation, some limitations were noted:

- *Slight color fringing near the disc edge due to minor chromatic aberration*
- *Manual focus control could introduce subjectivity; a digital focusing module can improve precision*
- *Current system doesn't support real-time solar data logging or image stacking*
- *Not Calibrated properly*

Future improvements may include:

- *Integration of an* ***auto-focus module***
- *High-resolution CMOS camera for live solar video*
- *Use of solar H-alpha filters for chromosphere imaging*
- *Better Calibration*

Conclusion

While the system demonstrated functional motion and was capable of following the general solar trajectory, ***automated tracking precision was not fully reliable*** *across all sessions. In each case,* ***manual intervention was necessary*** *to accurately center the Sun within the field of view. However, the progressive improvement observed between May and June confirms that the system responded well to iterative corrections in motor timing, mechanical tension, and alignment procedures.*

The ***3D-printed gear system*** *performed with acceptable stability but may have introduced slight backlash, which, combined with atmospheric effects and motor calibration limitations, reduced the accuracy of long-term automated tracking. Future improvements could include the integration of* ***closed-loop feedback (e.g., encoders or visual tracking)****, real-time correction algorithms, and temperature-compensated timing models for better celestial tracking.*

*Despite the limitations, the telescope provided **valuable solar imaging results**, and the current setup offers a solid testbed for further development in **low-cost automated solar instrumentation**.*

APPENDIX- I

SAMPLE CODE

```
#ifndef ASTRO_DATA_H

#define ASTRO_DATA_H

#include "Stepper.h"

#include <RTClib.h>

#include <Arduino.h>

#include <HTTPClient.h>

#include <WiFi.h>

#include <Wire.h>

RTC_DS3231 rtc;

int utcOffsetHours = 5;   // Change this to your local UTC offset (hours)

int utcOffsetMinutes = 30; // Change this to minutes offset (if applicable)

String objectName = "Sun"; // Default object

float temperature = 27.0, latitude = 22.0506247, longitude = 88.0717563, elevation = 16;

//float temperature = 27.0, latitude = 29.23, longitude = 79.50, elevation = 446;

String URL = "https://ssd.jpl.nasa.gov/api/horizons.api?format=json&COMMAND='";

bool API_status=false;

bool timeFound = false;

float Az = 0, preAz=0, Alt = 0, preAlt=0;

float apparentRA = 0, apparentDEC = 0;

String ra = "0h0ms", dec = "0°0'0\"";

struct objects {

   String object;

   int identifier;

};
```

```
objects data[] = {
    {"Sun", 10},
    {"Mercury", 199},
    {"Venus", 299},
    {"Mars", 499},
    {"Jupiter", 599},
    {"Saturn", 699},
    {"Uranus", 799},
    {"Neptune", 899},
    {"Pluto", 999},
    {"Moon", 301},
    {"Phobos", 401},
    {"Deimos", 402},
    {"Io", 501},
    {"Europa", 502},
    {"Ganymede", 503},
    {"Callisto", 504},
    {"Titan", 606},
    {"Enceladus", 610},
    {"Triton", 801},
    {"Ceres", 2000001},
    {"Vesta", 2000004},
    {"Eros", 2000433},
    {"Halley", 1000001},
    {"Bennu", 2101955},
    {"Voyager 1", 65531},
```

```
    {"Voyager 2", 65532},
    {"New Horizons", 65598},
    {"JWST", 65570}
};
uint16_t findObjectID(String name) {
    for (size_t i = 0; i < sizeof(data) / sizeof(data[0]); i++) {
        if (data[i].object.equals(name)) {
            return data[i].identifier;
        }
    }
    return 0; // Return 0 if not found
}
String fetchHorizonsData(String apiUrl) {
    HTTPClient http;
    http.begin(apiUrl);
    int httpCode = http.GET();
    String payload = "NA";
    if (httpCode == HTTP_CODE_OK)
    {
        payload = http.getString();
        API_status=true;
    }
    else
      API_status=false;
    http.end();
    return payload;
```

}

#endif

APPENDIX- II

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[11] Royal Observatory *of* Belgium

APPENDIX- III

Table of components with costing

Sl No.	Components	Quantity	Cost
1	Dobsonion Telescope 8 inch BRESSER MESSIER	1	48000
2	NEMA 17 stepper motor	1	1500
3	NEMA 23 stepper motor	1	1500
4	esp32 developer module	1	350
5	DRV8825 motor driver	2	220
6	Timing Belt 5mm	3 metre	240
7	Power Supply 24V 10A	1	2037
8	TMC2209 motor driver	2	2038

9	MPU9250	1	1568
10	NEO6MV2 GPS module	1	764
11	Magnetometer board HMC5883L	1	475
12	DS3231 RTC Module	1	423
13	Testing Gears(Pinion)	4	200
14	Wood cutting and drilling		600
15	Spiral wrapping band	4	120
16	esp32 Cable	1	70
17	Pulley	1	377
18	3D printing material (PLA plus)	1	1413
19	NEMA 23 stepper motor 30kgcm		2818
		Total--->	64713

APPENDIX- IV

ALL THE IMAGES TAKEN FROM THE TELESCOPE

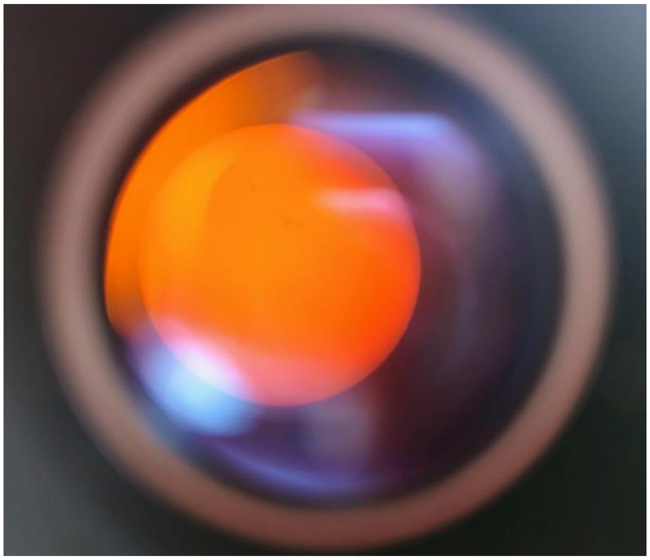

Fig 1: On 2nd May

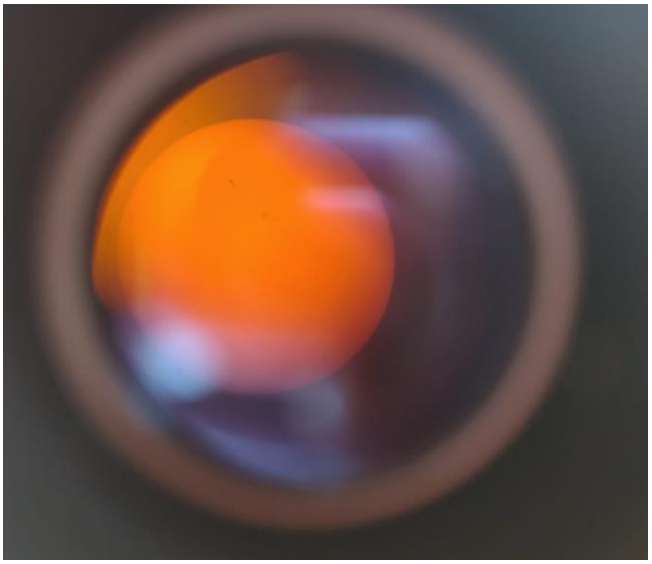

Fig 2: On 2nd May

Fig 3: On 2nd May

Fig 4: On 2nd May

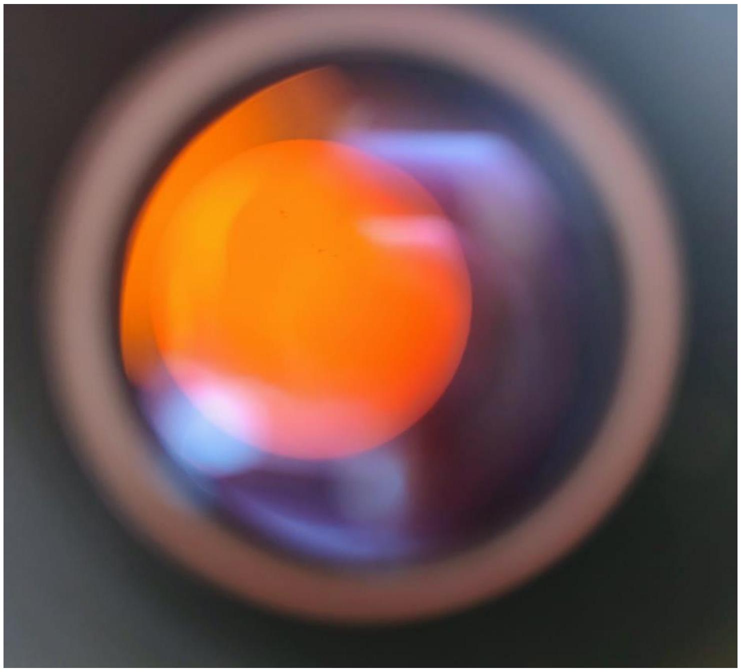

Fig 5: On 2nd May

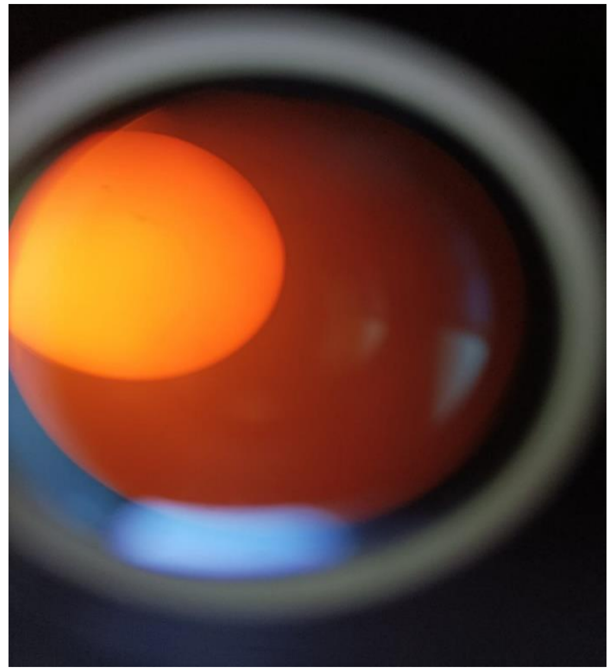

Fig 6: On 5th May

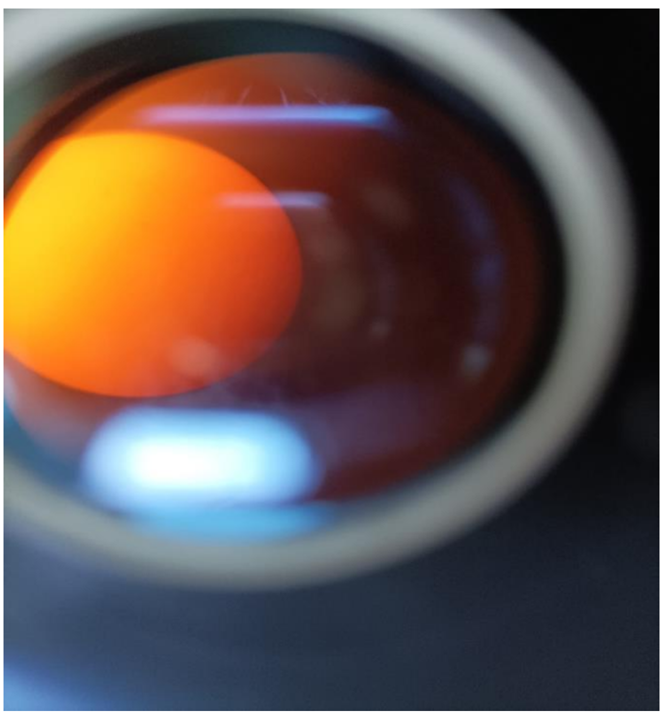

Fig 7: On 5th May

Fig 8: On 5th May

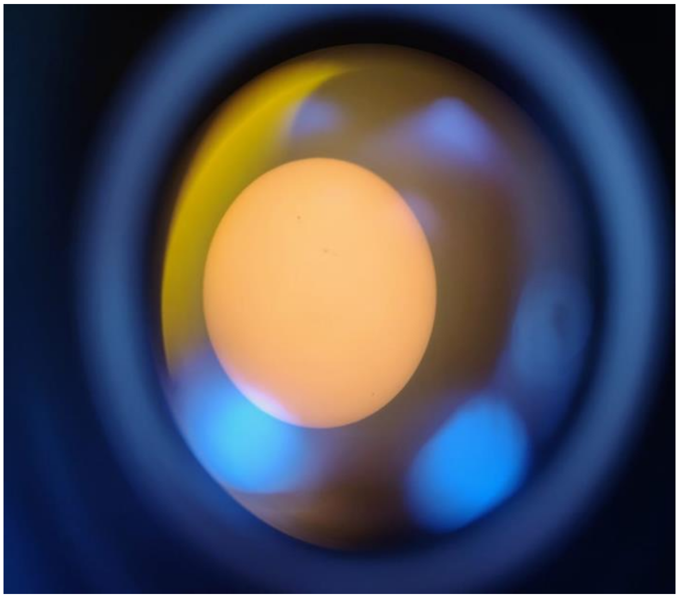

Fig 9: On 10th June 2025

Fig 10: On 10th June 2025

Fig 11: On 10th June 2025

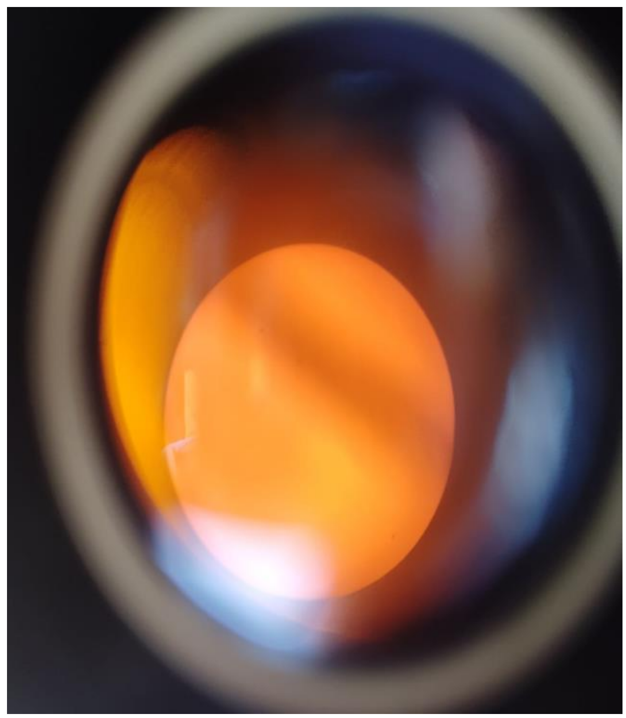

Fig 12: On 10th June 2025

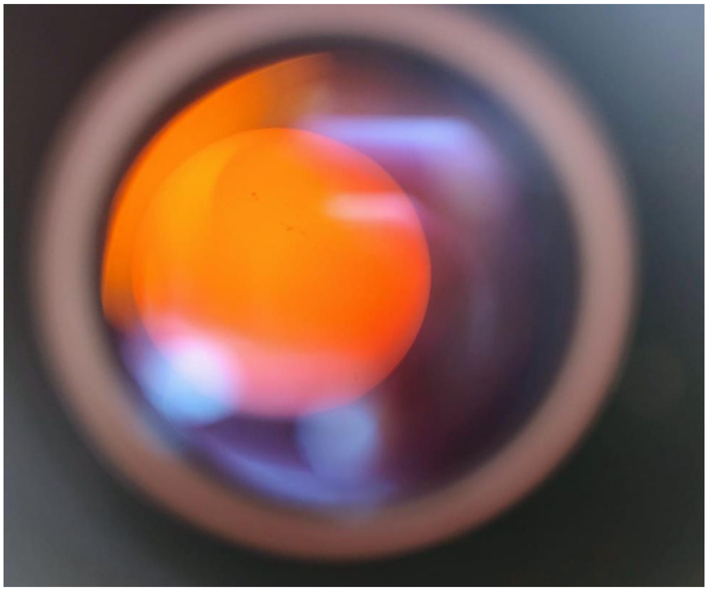

Fig 13: On 6th June 2025

Fig 14: On 6th June 2025

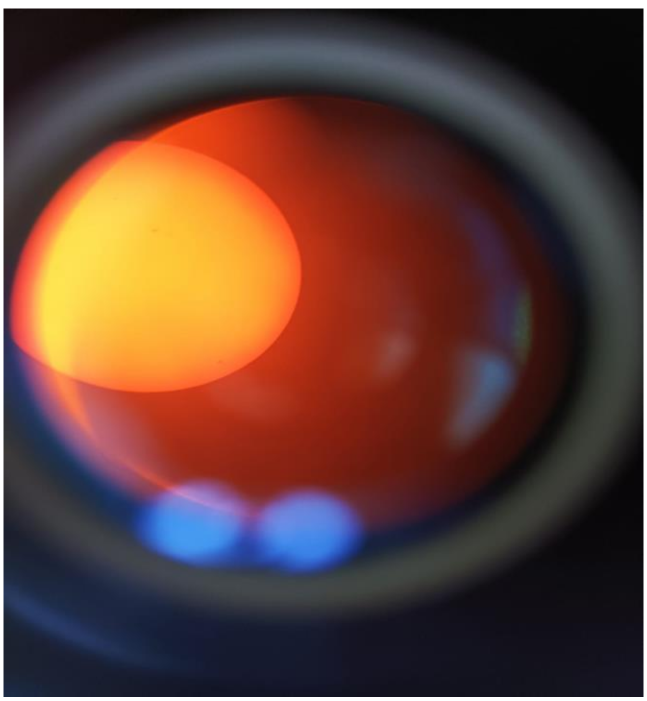

Fig 15: On 6th June 2025

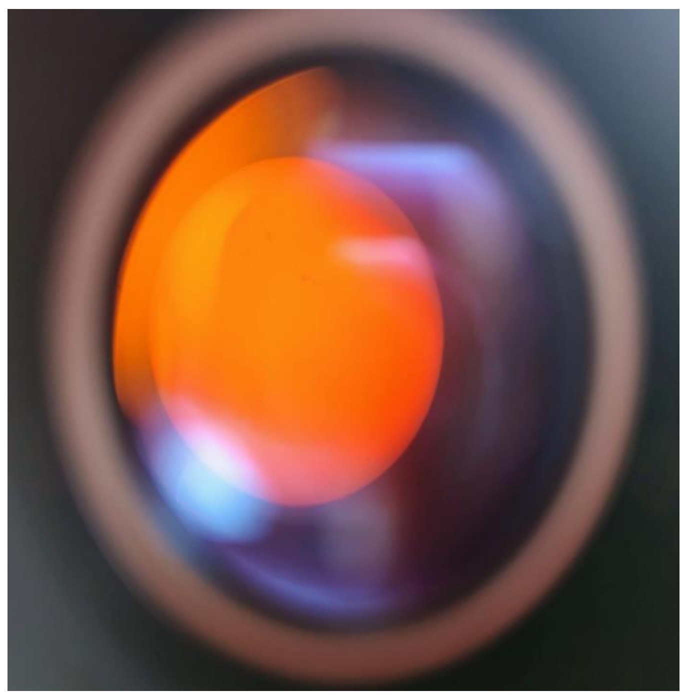

Fig 16: On 6th June 2025